



Project 3

University of Nebraska–Lincoln
Department of Civil Engineering
Engineering Analysis II

Team Members:

Christopher Moore

Owen Schafer

Mohamed Adam

Tom Steele

Instructor: Wissam Kontar

Section Number: 150

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Scope of Work

Project Title: Visualize Data

Team Members: Christopher Moore, Owen Schafer, Mohamed Adam, Tom Steele

Submitted to (Client): Federal Highway Administration (FHWA)

Project Goals:

The purpose of this project is to develop a Python-based code that will provide numerous detailed graphs. The graphs will be created using datasets provided by the National Household Travel Survey (NHTS) and Next Generation Simulation (NGSIM). The analysis will generate several visualizations to highlight key trends relevant to transportation planning.

Project Tasks:

Task 1: Create simple visual graphs

Task 1 will involve importing the provided CSV dataset and the associated metadata files into Python to generate several visual graphs. These visual graphs will include a bar chart, histogram, boxplot, and two time-series plots to illustrate important trends within the corresponding dataset.

Task 2: Simulation Study

Task 2 will implement Intelligent Driver Model (IDM) to simulate vehicle following behavior and trajectories.

Task 3: Timesheet

Task 3, utilizing a time sheet, will allow the client to understand how the project team used the allocated time towards analysis and creation in Python. The provided time sheet will specifically provide information to the client to evaluate the efficiency of Python compared to Excel for data analysis tasks.

Task 4: Create Deliverables

Task 4 includes the creation of a GitHub link, Python code, SOW, Gantt chart, time sheet, annotated document code, and a finalized report. This will establish a professional submission to summarize the project findings.

Summary:

Introduction:

Transportation systems are used every day, ranging from personal travel to economic and professional activities. To better understand how people travel and how traffic behaves, our client, Federal Highway Administration (FHWA), tasked our team with transforming provided datasets into clear and professional visualizations for analysis. The National Household Travel Survey (NHTS) holds data focused on how people travel, which includes trip purpose, distance, and mode of transportation. While the NGSIM dataset focuses on driving behavior, which includes the vehicle speed and movement over time. The goal of this project was to create a python code that will organize, visualize, and analyze provided datasets. Different graphs have been created to show important trends of interest to our client. In addition, a simulation using the Intelligent Driver Model has been used to model how vehicles follow each other on the road.

Methods, Results, and Conclusion:

The data was imported into Python using pandas and was cleaned afterwards. Any unnecessary values were removed and only variables relevant to the analysis were kept. The variables chosen were Vehicle Type, Vehicle Age, Household Location (Urban vs Rural), Fuel Type, and Vehicle Make. These variables are a good indicator of how people across the United States behave when it comes to vehicle usage. We used different types of graphs to analyze this data. Bar charts helped us to compare variables like the vehicle type and fuel type. We used histograms to show the distribution of values that are numbered like the age of the vehicles. Boxplots compared distributions of the vehicle age across categories like the vehicle make or household location. To generate these plots in python we had to use the matplotlib and seaborn libraries as well as add labels, titles, and other things to improve the graph.

Variables for the NGSIM dataset relating to driving behavior were selected. We chose to use speed, acceleration, position, time, and gap distance between vehicles. These variables are important to understanding how vehicles interact with each other on the road. From these variables we created a time series plot showing a speed vs time and acceleration vs time graph, a position vs time plot to observe vehicle trajectories, a gap

distance plot to show the distance between vehicles on the road, and a histogram to show the acceleration distribution. These graphs identify patterns between vehicles on the road and show human driving behavior like how acceleration changes, speed and spacing fluctuations, and more.

To analyze the behavior of vehicles better we used a simulation produced by the Intelligent Driver Model (IDM). This model predicts how a driver would operate a vehicle and simulates how a driver would accelerate or brake based on current speed, gap distance, desired speed, and a following distance. We used a leader-follower pair to simulate this. The leader's trajectory data was used as input and the follower's behavior was simulated using the IDM equation. This simulation helped to calculate the follower acceleration over time and the position and speed of the simulated vehicle.

The findings from this project can be applied to real world transportation engineering projects, planning, and future policy decisions. Understanding the types of vehicles Americans use can help with infrastructure design. If more electric vehicles were in use, gas stations would be replaced with charging stations. Analyzing driving behavior through the dataset also can help develop safer roads and improved traffic flow models.

This project used Python to analyze and visualize the transportation data from the NHTS and NGSIM datasets. From the visualizations, it was clear that personal vehicles are the most popular form of transportation and traditional gas-powered vehicles will continue to be in use. Differences between urban and rural areas help future engineers understand the transportation needs each location requires. Also, it showed how driving behavior is not linear but highly dynamic and constantly changing speed, acceleration, and reacting to situations. The IDM model demonstrates how models can be used to approximate behavior but it also shows how models are not entirely accurate at predicting all of a human's behavior. Overall this project will allow future engineers to better understand traffic patterns and improve roadway safety.

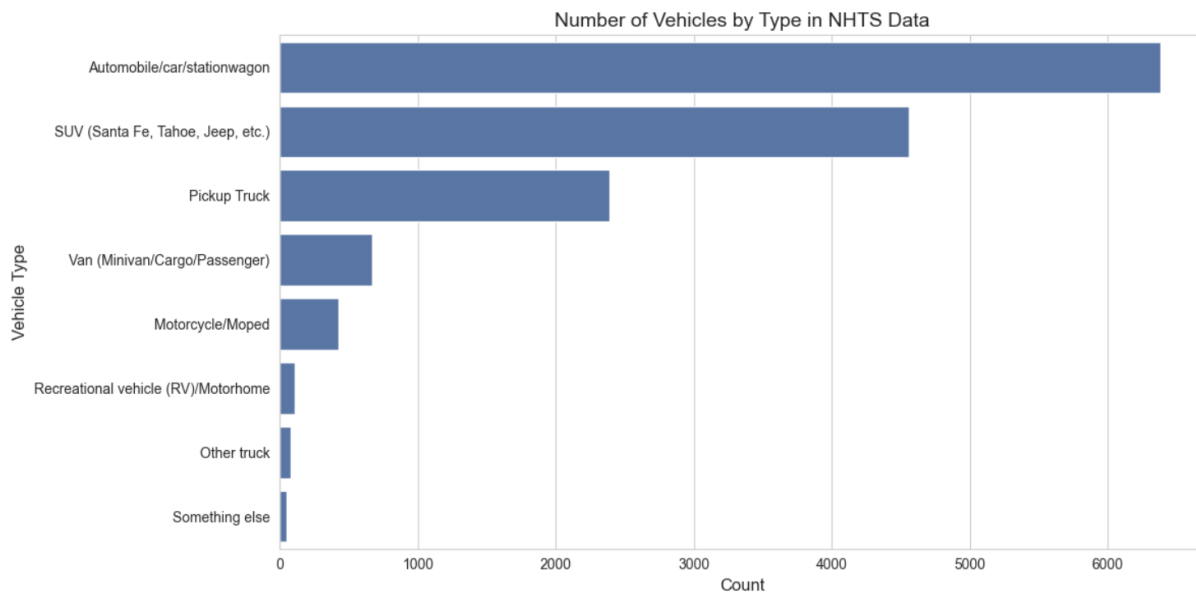


Figure 1. Vehicle Type vs. Count – NHTS Dataset

Figure 1 is a bar chart that shows the number of vehicles by type in the NHTS dataset. Based on this data we can see automobiles, cars, and station wagons are the highest number of vehicle types. SUVs and Pickup trucks combined are a little more than the automobiles, cars, and station wagons. Passenger vehicles are the most common which show a strong reliance on personal vehicles for common citizens in the United States. Compared to the low number of vans, motorists, and recreational vehicles this indicates to transportation planners that the US will continue to have needs for improved roadway infrastructure for personal use.

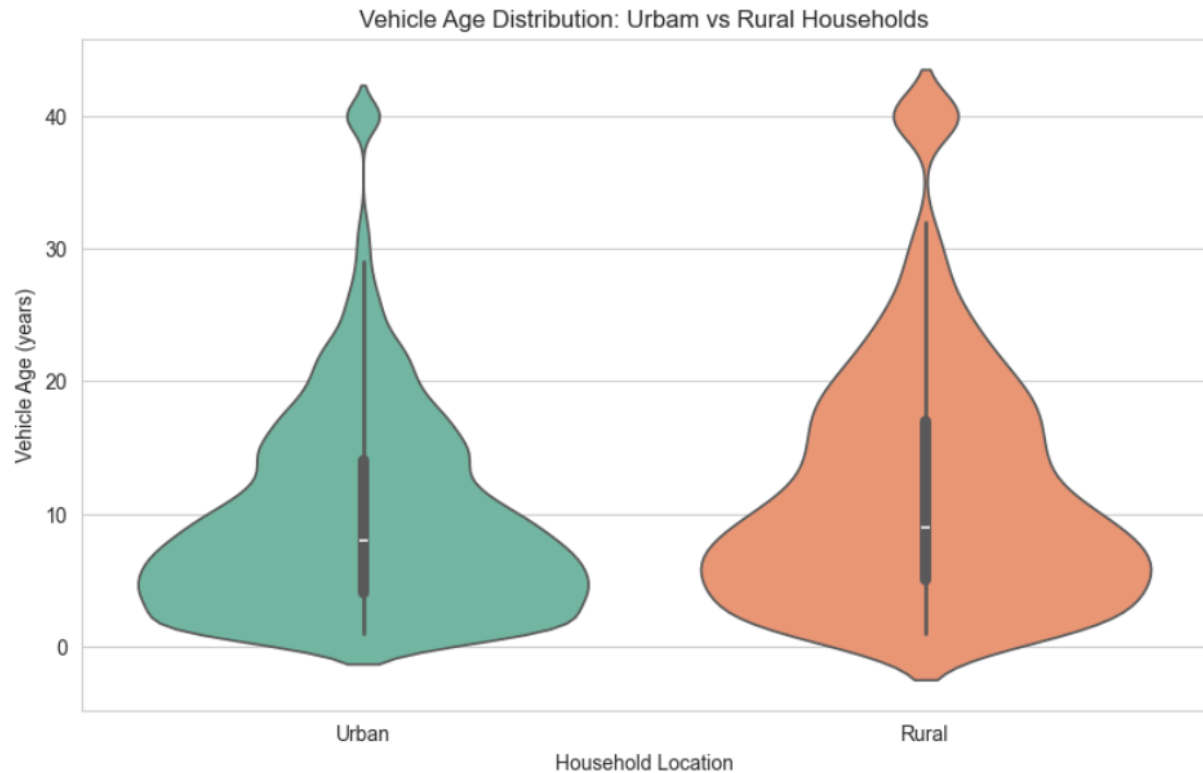


Figure 2. Vehicle Age (years) vs. Household Location - NHTS Dataset

This vehicle age distribution graph shows that most vehicles fall within an early age range compared to older and very older vehicles. Vehicles in both urban and rural areas are far more likely to be under the 0-20 year age range. Americans are more inclined to newer vehicles and this indicates that more safety regulations, maintenance needs, and roadway infrastructure needs attention to keep the streets safe and moving well. As Americans improve for the future transportation planners and engineers will be right there with them.

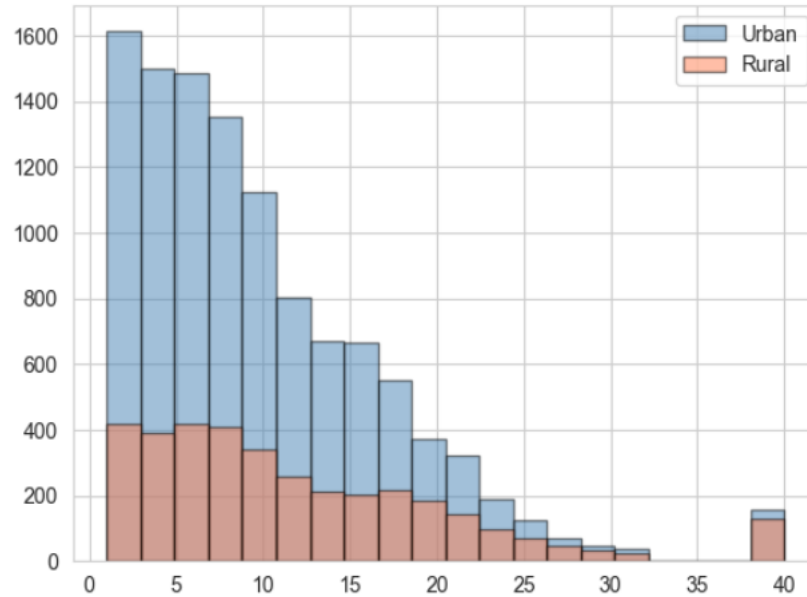


Figure 3. Bar chart Urban and Rural - NHTS Dataset

This bar chart compares urban and rural households in the dataset. There is a far greater population of urban homes but rural houses are also plentiful. Transportation policies need to be geared to consider the needs and safety of both densely and less populated areas.

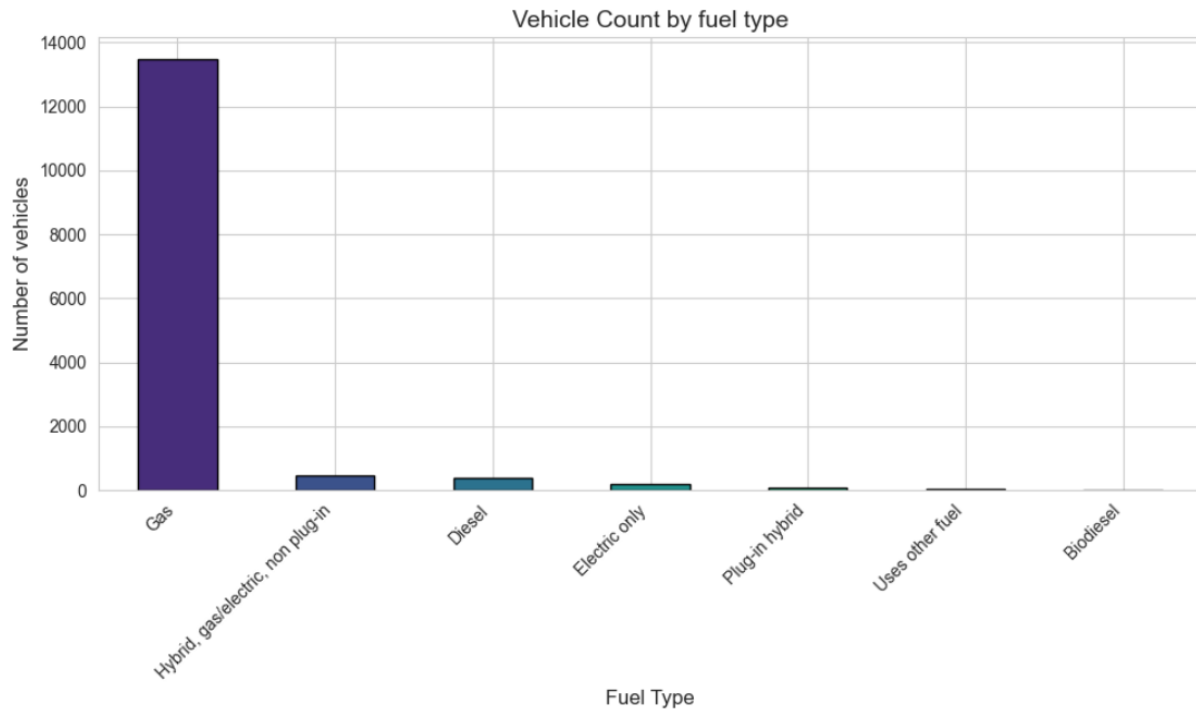


Figure 4. Number of Vehicles vs. Fuel Type – NHTS Dataset

Figure 4's chart shows the distribution of fuel types across vehicles. Traditional gasoline-powered vehicles dominate the dataset. Alternatives like hybrids, electric, and diesel-powered vehicles also exist but the vast majority of cars rely on gasoline. Sustainable and environment safe transportation is growing but at a very slow pace so for the time being it has not replaced conventional fuel sources. This will slow down the growth of environmental policy and future infrastructure planning.

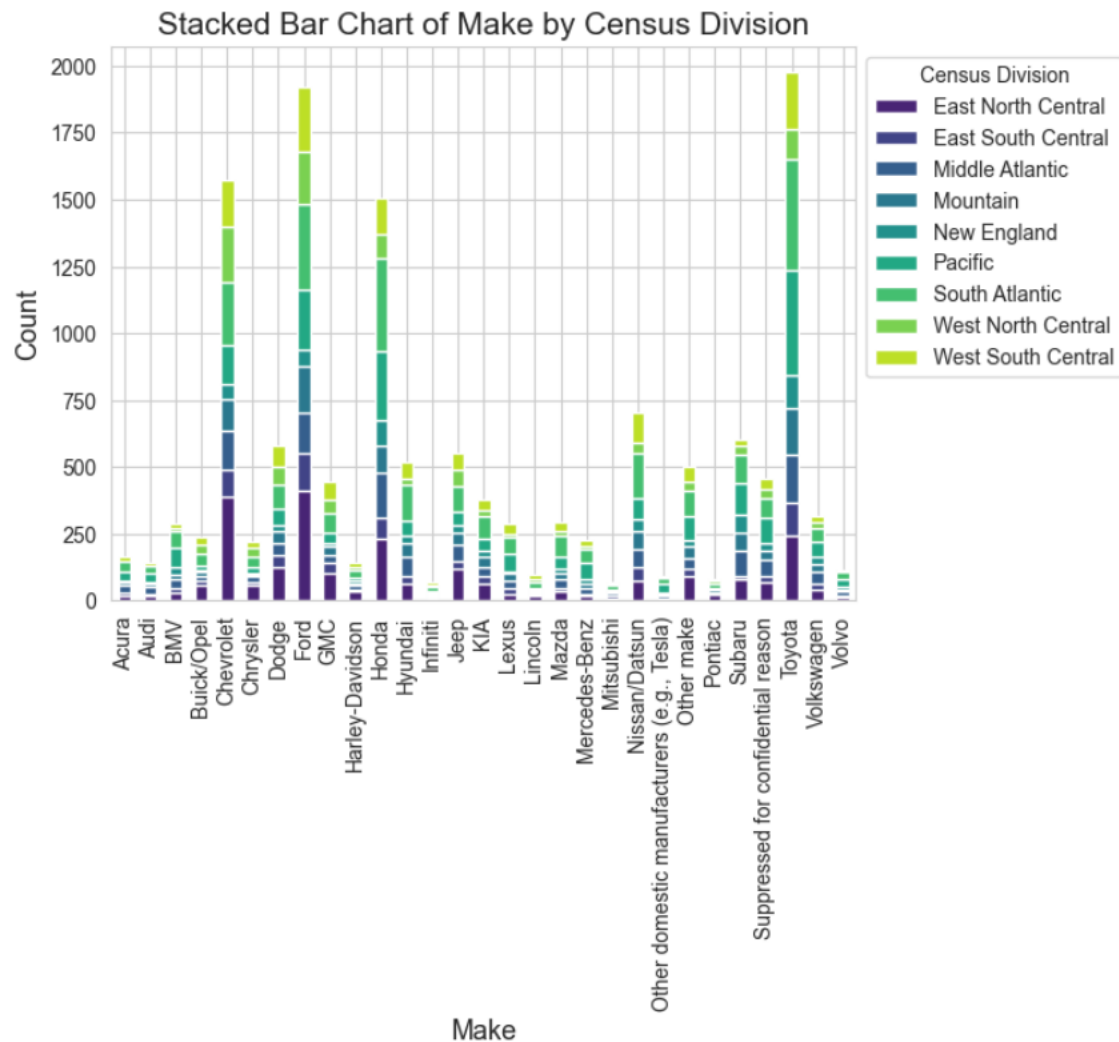


Figure 5. Count vs. Make - NHTS Dataset – Bar Chart

Figure 5 shows which vehicle manufacturers are most common and the areas across the US in which they are most common. The most popular brands are Chevrolet, Ford, Honda, and Toyota. These brands indicate consumer preference & market dominance and this information is useful to understanding vehicle reliability trends and purchasing patterns.

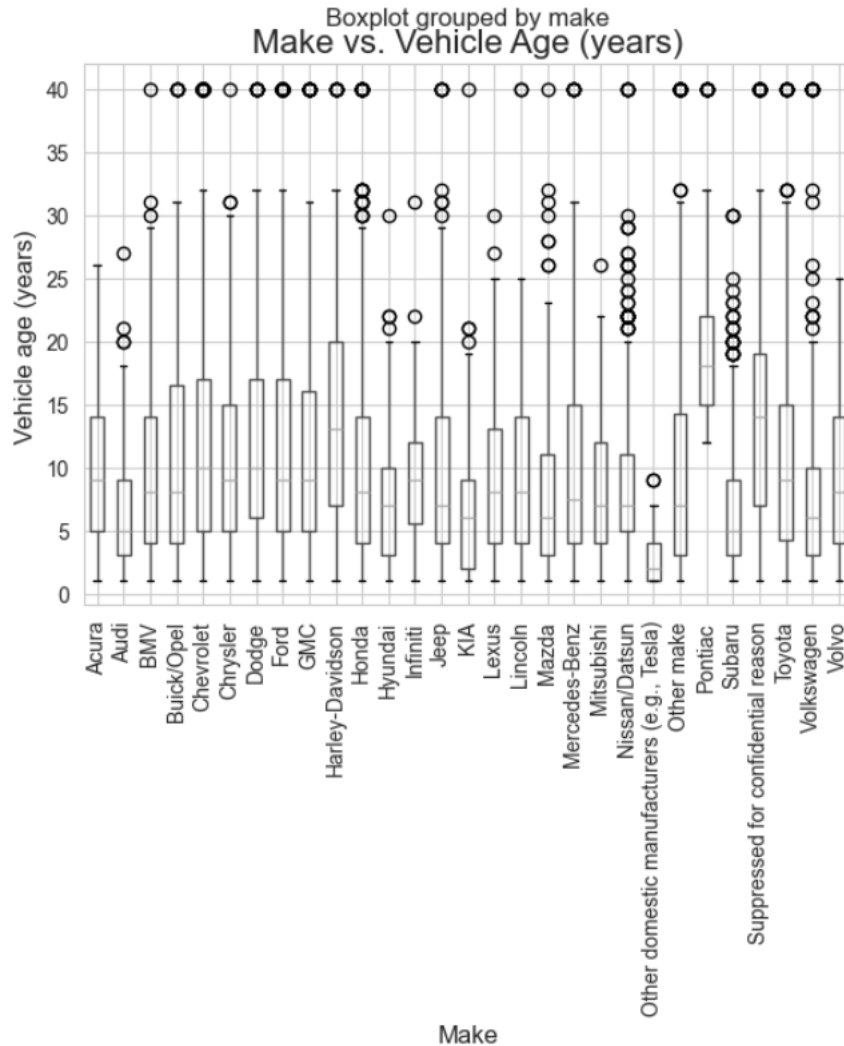


Figure 6. Vehicle Age (Years) vs. Make – NHTS Dataset

This boxplot shows how vehicle age varies across different manufacturers. Some brands have a wider spread in age which means their products stay in use longer. Others have a tighter range which indicates quicker replacement cycles. This means differences in durability, consumer behavior, or resale value.

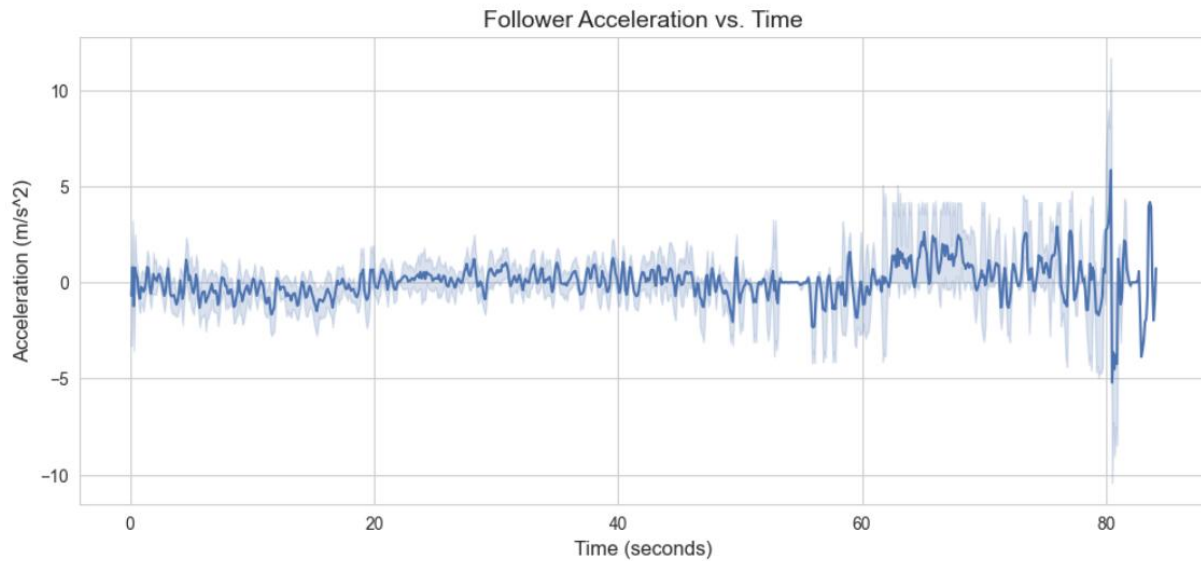


Figure 7. Acceleration (m/s²) vs. Time (Seconds) – NGSIM Dataset

This time-series plot shows the follower acceleration by time. Most values stay close to zero with some small spikes. This makes sense given normal driving behaviors include drivers making small adjustments. Large spikes indicate the drivers may have braked suddenly or accelerated, likely due to congestion or a reaction.

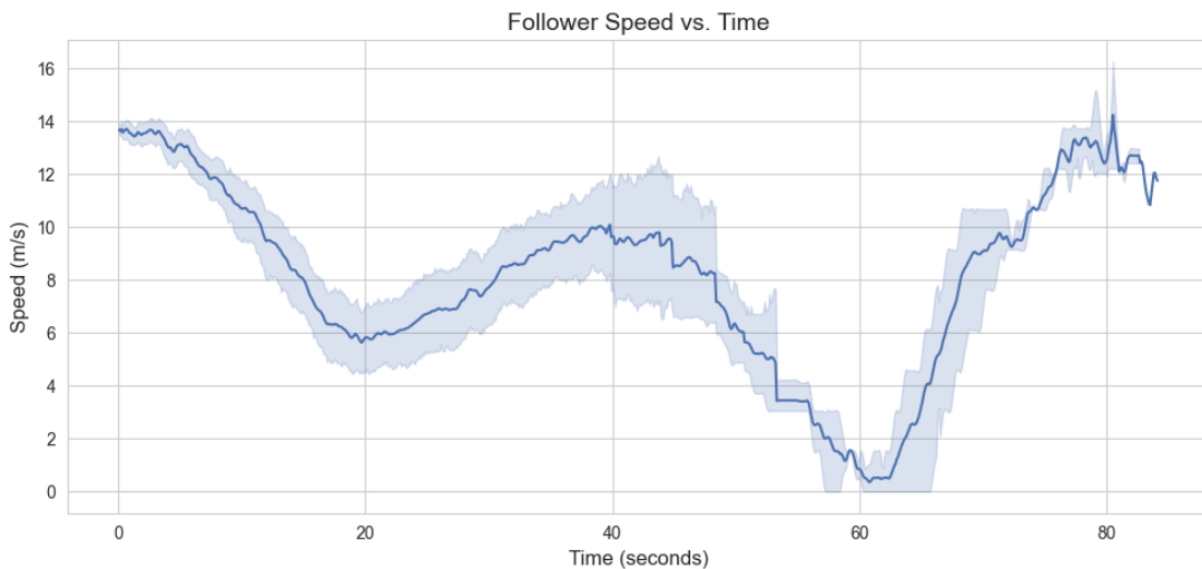


Figure 8. Speed (m/s) vs. Time (seconds) – NGSIM Dataset

This plot shows the follower speed vs time. The speed is steady and then drops followed by another increase. It repeats like a parabolic graph in both directions. This suggests traffic flow isn't constant and is affected by surrounding vehicles.

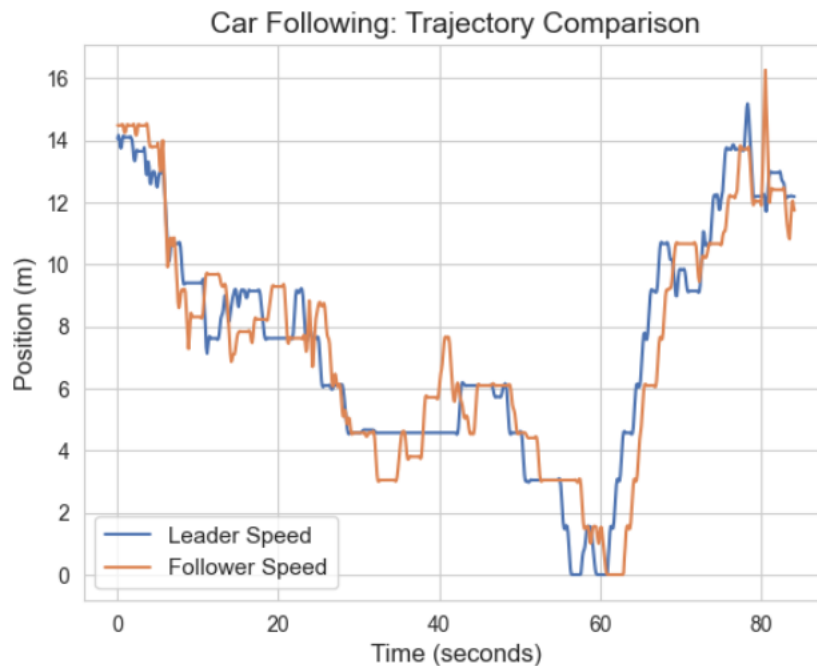


Figure 9. Position (m) vs. Time (seconds) – Speed Comparison - NGSIM Dataset

This graph shows the trajectory comparison between both leader and follower vehicles. The position changes over time for different vehicles. The slope of each line represents speed. The differences in each line show variations in driving speed between the vehicles.

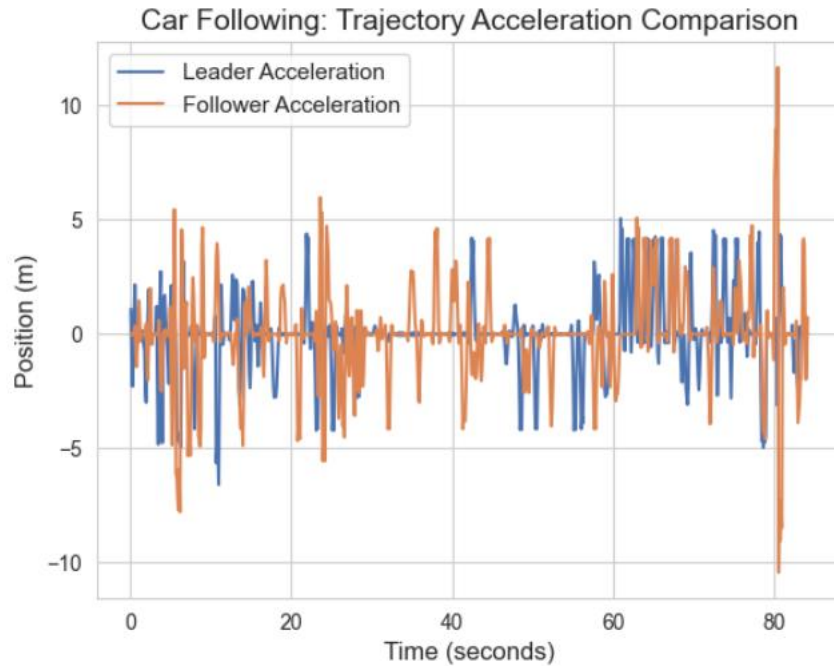


Figure 10. Position (m) vs. Time (seconds)- Acceleration Comparison - NGSIM Dataset

This graph compares the acceleration and how it affects vehicle position. Sudden changes show where the acceleration or deceleration occurs. For example, at 80 seconds the acceleration change of the follower is much higher compared to the acceleration of the leader.

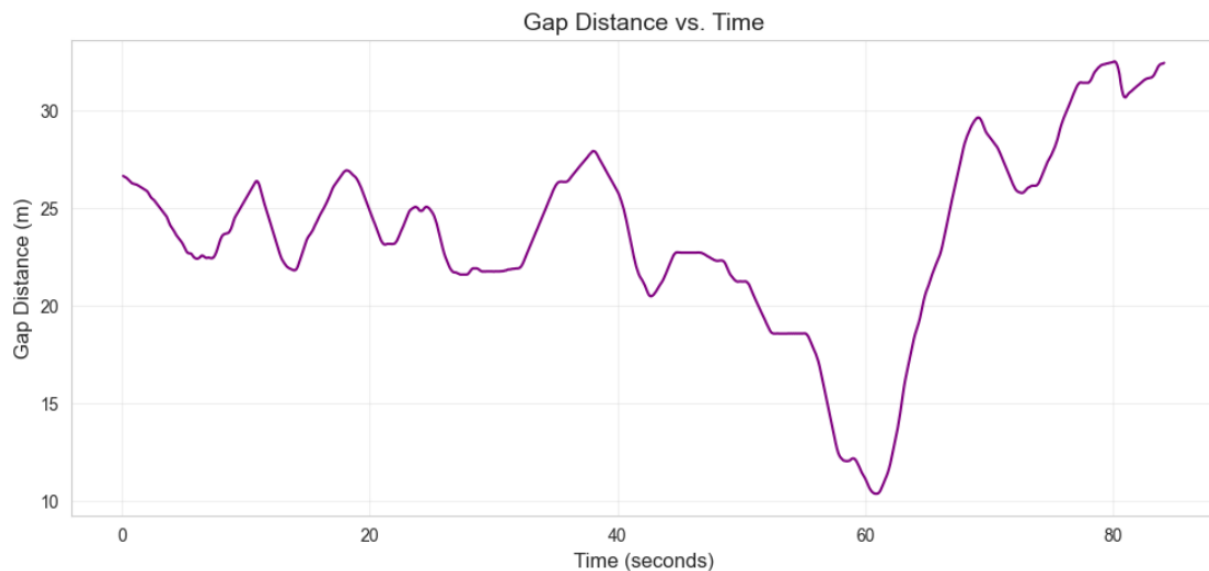


Figure 11. Gap Distance (m) vs. Time (seconds) - NGSIM Dataset

Figure 11 compares the gap distance of the vehicles over time. Smaller gaps can indicate aggressive driving such as tailgating, break-checking, or driving between vehicles without a safe following distance. Large gaps indicate safe following distance from the follower vehicle and an understanding of safety laws.

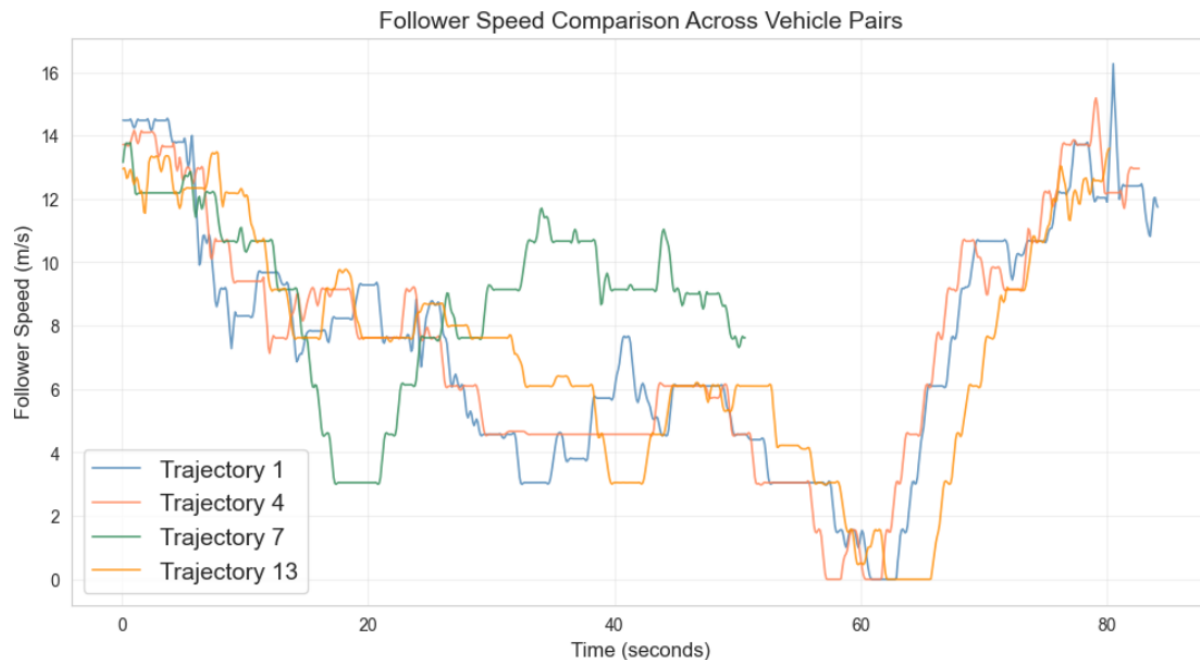


Figure 12. Follower Speed (m/s) vs. Time (seconds) – NGSIM Dataset

This plot focuses on the follower speed across vehicle pairs. The speed changes are usually the same in the beginning 15 seconds and last 60 to 80 seconds but vary a lot in between. Drivers react to the vehicle in front of them.

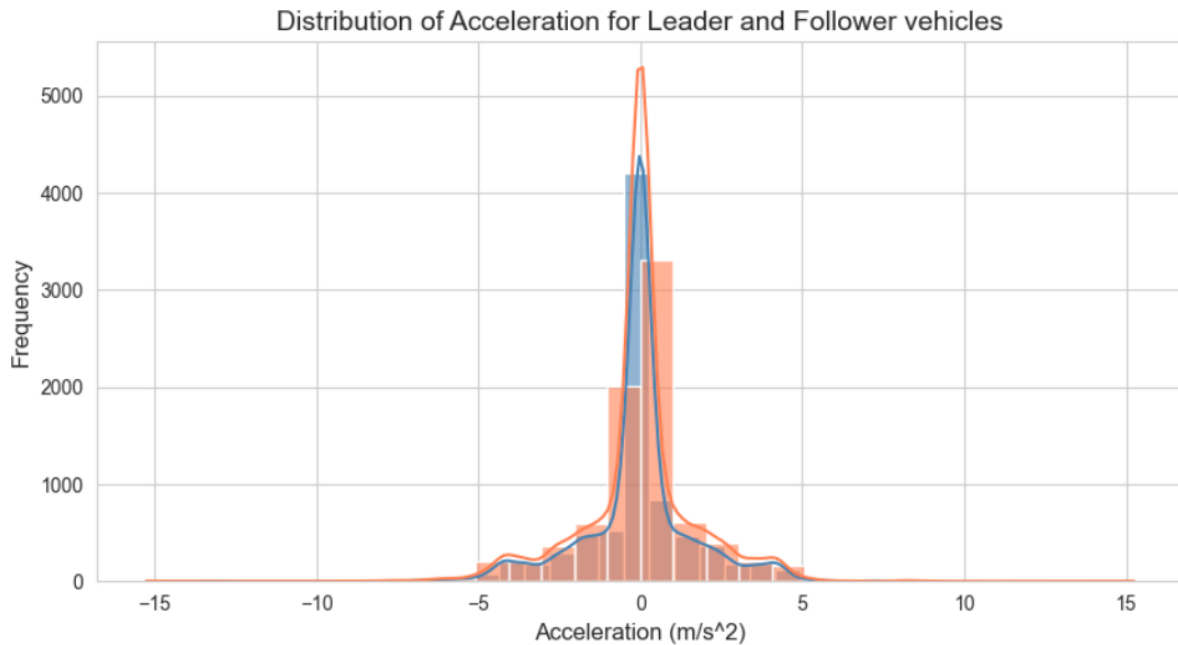


Figure 13. Frequency vs. Acceleration (m/s^2) – NGSIM Dataset

This is a graph highlights the distribution of acceleration for leader and follower vehicles. Most acceleration values are clustered near 0 with some at the range from -5 to 0 and 0 to 5. Vehicles spend most of their time maintaining speed rather than accelerating hard and braking hard. Extreme values are probably reactions to sudden changes in traffic.

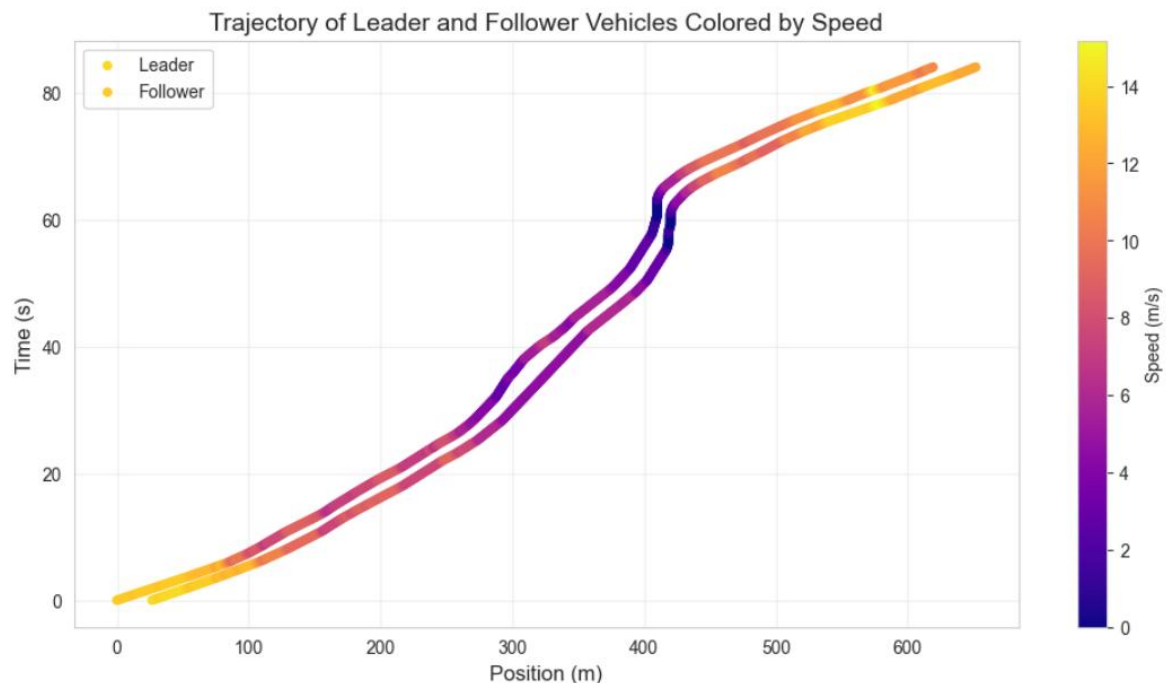


Figure 14. Time(s) vs. Position (m) – NGSIM Dataset

This plot shows the trajectory of vehicle movement over time. The color patterns also indicate speed with lighter colors being faster and darker colors being slower. The trajectories seem to match.

Intelligent Driver Model (IDM):

Acceleration:

$$\frac{dv}{dt} = a \left[1 - \left(\frac{v}{v_0} \right) - \left(\frac{S^*(v, \Delta v)}{s} \right)^2 \right]$$

Desired minimum gap:

$$s^*(v, \Delta v) = S_0 + vT + \frac{v\Delta v}{2\sqrt{ab}}$$

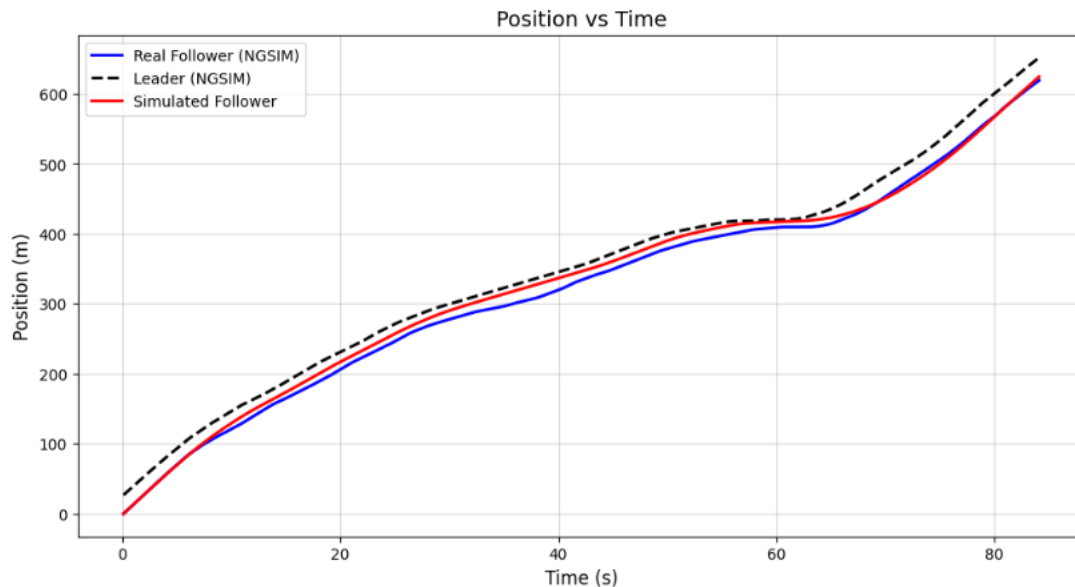


Figure 15. Position (m) vs. Time (s) – NGSIM Dataset

This figure compares the simulated follower vehicle to the real follower data while also showing the leader data. The trend is similar which means an IDM model is effective at predicting behavior. There are small differences that show how human error and reactions and unpredictability cannot be shown perfectly by a model.

References:

Data sources:

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